

Axona Quick Start

Get a working pub/sub roundtrip in **under five minutes**. Two Node peers, one process, no browser, no bridge. Once it works you've verified your setup; jump to the Programmer Guide for the full picture and the API Reference for every exported symbol.

Prerequisites

- **Node.js 20+** (for built-in Web Crypto Ed25519)
- A terminal

That's it. No build step, no DB, no bridge.

1. Install (30 seconds)

```
mkdir my-axona-demo && cd my-axona-demo
npm init -y
npm pkg set type=module
npm install github:axona-net/axona-protocol#v2.40.0
```

2. Write the demo (one file)

Save this as `index.js`:

```
import {
  AxonaPeer, AxonaDomain, NeuronNode, AxonaManager, Synapse,
  SimNetwork, simTransport,
  deriveIdentity, geoCellId, clz264,
} from '@axona/protocol';

// Region: us-east (Virginia). Both peers live here.
const REGION = { lat: 38.0, lng: -77.0 };

// Synthetic publisher whose top 8 bits = the region's S2 cell, so
// the derived topic id has matching S2 prefix → us-east peers naturally
// cluster as the axon set.
function regionSynthPublisher({ lat, lng }) {
  const s2 = geoCellId(lat, lng, 8);
  return s2.toString(16).padStart(2, '0') + '0'.repeat(64);
}

// Build a peer from kernel primitives.
async function makePeer(network) {
  const identity = await deriveIdentity(REGION);
  const transport = simTransport({ network, identity, heartbeatMs: 0 });
  await transport.start(identity.id);

  const node = new NeuronNode({
    id: BigInt('0x' + identity.id), // NeuronNode XORs IDs as BigInts
    lat: REGION.lat, lng: REGION.lng,
  });
  node.transport = transport;
  const domain = new AxonaDomain({ k: 20 });
  const peer = new AxonaPeer({ domain, node, identity, transport });
  await peer.start();
}
```

```

// Wire AxonaManager (pub/sub layer) to AxonaPeer's DHT primitives.
const dht = {
  getSelfId:      () => peer.getNodeId(),
  findKClosest:   (...a) => peer.findKClosest(...a),
  routeMessage:   (...a) => peer.routeMessage(...a),
  sendDirect:     async (peerId, type, payload) => {
    if (peerId === peer.getNodeId()) { // self-loop
      const h = peer._directHandlers?.get(type);
      if (!h) return false;
      await h(payload, { fromId: peer.getNodeId(), type });
      return true;
    }
    return peer.sendDirect(peerId, type, payload);
  },
  onRoutedMessage: (t, h) => peer.onRoutedMessage(t, h),
  onDirectMessage: (t, h) => peer.onDirectMessage(t, h),
};
peer._axonaManager = new AxonaManager({ dht });
return { peer, identity };
}

// Two peers on a shared SimNetwork.
const network = new SimNetwork();
const alice = await makePeer(network);
const bob    = await makePeer(network);

// Open a channel between them and admit each as a synapse (real
// transports do this via the axona:hello handshake on channel-open).
await alice.peer._transport.openConnection(bob.identity.id);
const admit = (local, remoteHex) => {
  const remote = BigInt('0x' + remoteHex);
  const syn = new Synapse({
    peerId: remote, latencyMs: 1,
    stratum: clz264(local._node.id ^ remote),
  });
  syn.weight = 0.5; syn.inertia = 0; syn._addedBy = 'demo';
  local._node.synaptome.set(remote, syn);
};
admit(alice.peer, bob.identity.id);
admit(bob.peer,   alice.identity.id);

// Pub/sub roundtrip.
const TOPIC    = 'us-east/hello-world';
const publisher = regionSynthPublisher(REGION);

const sub = await bob.peer.sub(TOPIC, (env) => {
  console.log('[bob] received:', env.message);
}, { publisher, since: 'all' });

await new Promise(r => setTimeout(r, 100)); // let subscribe-k land

const msgId = await alice.peer.pub(TOPIC, 'hello from alice', { publisher });
console.log('[alice] published msgId=' + msgId);

```

```
await new Promise(r => setTimeout(r, 200)); // let fan-out complete
await sub.stop();
process.exit(0);
```

3. Run

node index.js

You should see:

```
[alice] published msgId=8e9d4b1a...
[ bob ] received: hello from alice
```

That's it. You just ran:

- Ed25519 identity derivation (264-bit nodeIds, S2-cell-prefixed for us-east)
- A signed envelope publish replicated to the topic's $R = 5$ root axons (the root-set size — distinct from the Kademlia routing parameter $k = 20$)
- A `since: 'all'` subscriber receiving the cached message

The full version of this with comments and a verifier is at `examples/minimal-pubsub/` — same code, ready to run.

What just happened

alice	bob
<code>peer.pub(TOPIC, ...)</code>	<code>peer.sub(TOPIC, handler, since:'all')</code>
AxonaManager derive 264-bit topic id from the synth publisher build signed envelope send 'publish-k' to K-closest	AxonaManager derive same topic id (same publisher arg → same id) send 'subscribe-k' to K-closest (alice + bob - small mesh)
alice cache role	bob cache role (lazy-promoted on first publish-k) bob is a child of its own role
	<code>deliveryCallback(env) handler</code>

Both peers compute identical 264-bit topic ids because they use the same `opts.publisher` value (the synthetic region id). That ID-matching is the **one rule you can't break** — see Pitfall #13.1 in the Programmer Guide.

Next steps

Want to...	Read
Understand the mental model	Programmer Guide §3
Build a chat / forum / feed app	Programmer Guide §12
Hook up a real WebRTC + bridge stack	<code>axona-peer/src/axona_node.js</code> — the reference browser wiring
Look up a specific function	API Reference
Run a bridge locally	Programmer Guide §11

Troubleshooting

Cannot find module '@axona/protocol' — make sure `package.json` has `"type": "module"` and the install completed. Re-run `npm install`.

Cannot mix BigInt and other types — your `NeuronNode` got an id of the wrong type. Pass `BigInt('0x' + identity.id)`, not the raw hex string.

Demo runs but nothing arrives at bob — the synaptome admission step is missing. Real transports admit peers via `axona:hello`; in this in-process demo you have to do it manually (the `admit(...)` calls above).

UPGRADE_REQUIRED close code (4426) — only happens when connecting to a bridge; not relevant for the in-process demo. If you see it when expanding to a bridge connection, your peer is older than the bridge's `MIN_PEER_VERSION`. Update both to `v2.10.0+`.